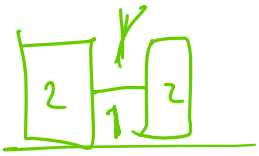
Lec26

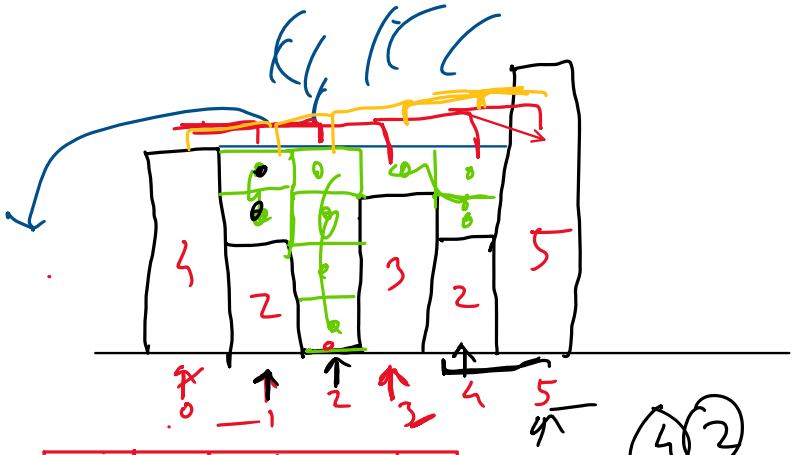
11 March 2026 02:57 PM

Trapping rain water

arr = [4, 2, 0, 3, 2, 5]

0 1 2 3 4 5





4 5

4 - 0 = 4

4 5

4 - 2 = 2

Left = | 4 | 4 | 4 | 4 | 4 | 5 |

0 1 2 3 4 5

i i i i i i

4 2

4 0

i = 0

Right = | 5 | 5 | 5 | 5 | 5 | 5 |

i i i i i i

i+1

= | 4 | 2 | 0 | 3 | 2 | 5 |

i i i i i i

4, 5 = 4 - 4 = 0

4, 5 = 4 - 2 = 2

4, 5 = 4 - 0 = 4

4, 5 = 4 - 3 = 1

4, 5 = 4 - 2 = 2

5, 5 = 5 - 5 = 0

Himanshu

100 →

101 →

arr = [4, 2, 0, 3, 2, 5] , n = arr.length

```
Left = new int[n]
Left[0] = arr[0]
for (i = 1; i < n; i++){
    Left[i] = max(Left[i-1], arr[i])
```

```
for (i = 1; i < n; i++){
    Left[i] =     max (Left[i-1], arr[i])
}
Right = new int[n]
Right[n-1] = arr[n-1]
for (i = n-2; i >= 0; i--){
    Right[i] = max(Right[i+1], arr[i])
}

totalwater = 0
for (j = 0; j < n; j++){
    totalwater += min (Left[j], Right[j]) - arr[i];
}
return totalwater
```